

Analysis of Six-Pulse Bridge Circuits Including Zener and Semiconductor Diodes Part One

Mohamed M. Saied*

Abstract

This paper addresses the analysis of electronic bridge circuits including Zener and/or semiconductor diodes. The analysis is based on recently suggested models for these diodes. A compact Mathematica code written in the Versions 7.0, 11.3 and 12.2 simulating a six-pulse rectifier bridge circuit is then derived and mathematically implemented. It investigates the time-domain transient and steady-state analysis of several representative case studies. In principle, the computer simulation can yield the circuits' currents and voltages as functions of time resulting from voltage sources of any waveform. Special emphasis is given, however, to power frequency sinusoidal sources. The maximum phase value of the 60-Hz source is assumed 12 Volts. The eventual presence of load at the bridge output terminals as well as the effect of the smoothing filter circuits are taken into consideration. The investigation of the resulting valve stresses can be of crucial importance when discussing the usability of Zener diodes as building elements in bridge circuits. The source currents are relatively smooth curves oscillating between -3.5 and +3.5A and exhibiting small high frequency as well as noticeable DC offset components. It will be shown that the models dealing with semiconductor diodes can be easily derived from those addressing Zener diodes, through simple modifications.

Keywords: Six-pulse, Diodes, Semiconductor devices, Transients, Mathematical model, Circuit analysis, Simulation, Electronic circuits, Bridge Circuits, Rectifiers, *Mathematica*, Zener

INTRODUCTION

The semiconductor diodes have been traditionally used as the main building blocks of rectifier circuits. This trend started from the early simple basic circuits to the more sophisticated connections such as the four- and six-pulse rectifier bridge circuits. [1-2] As mentioned in one of the interesting discussions,[3], high maximum peak reverse voltages are expected in rectifier circuits. This is the main reason why Zener diodes are not used for rectification purposes but are mostly used in applications that require voltage regulation. The aim of the publications [4-6] was to get an explanation why the use of Zener diodes in full-wave rectifier circuits is not suitable.

In summary, the feasibility of using these diodes in rectifier circuits are still not clear. It should be determined by the diode stresses in addition to the assessment of the economical aspects. This justifies the need for a detailed and egregious analysis of practical circuits equipped with this kind of diodes. This paper is a step in that direction.

*Author for Correspondence

Mohamed M. Saied*
 E-mail: mmsaied2@yahoo.com

Senior Member, IEEE Professor (Emeritus), Department of Electrical Engineering, Kuwait University, Independent Researcher, Giza, Cairo, Egypt

Received Date: April 12, 2022
 Accepted Date: May 17, 2022
 Published Date: May 25, 2022

Citation: Mohamed M. Saied. Analysis of Six-Pulse Bridge Circuits Including Zener and Semiconductor Diodes. Journal of VLSI Design Tools & Technology 2022; 12(1): 14–22p.

As a representative example, the main objective of this investigation is the derivation and implementation of a flexible and user-friendly model for simulating six-pulse circuits. Several recent versions of the software *Mathematica* have been selected for writing the corresponding computer code. As will be seen, the approach is capable of dealing with the transient processes as well as the steady state conditions of the circuit in

response to any wave forms of the applied source voltages. Furthermore, the 3-phase voltages are not necessarily symmetrical and the six diodes can be of different technical data. It will be further shown that the model can handle the presence of diodes of both semiconductor and/or Zener types and bridge circuits with faulty diodes.

METHOD OF ANALYSIS

The resistance of the Zener diode can be expressed as a function of the voltage v across the diode by the following simple equation:

$$R = rf + (rr - rf) \cdot (u[v + V_Z] - u[v]). \quad (1)$$

In which $u[x]$ denotes the unit-step function. Figure.1 shows the circuit diagram of a six-pulse rectifier bridge connected to a 3-phase voltage source having the phase voltages $e_a(t), e_b(t), e_c(t)$. The source has the internal resistance R_s and inductance L_s .

The voltage-current characteristic and the dependence of the diode's resistance of the The its voltage are illustrated by the two figures Figure 2 and Figure 3, respectively. The bridge is loaded by the load of resistance R_{load} . The element C represents the eventually existing shunt- connected smoothing capacitor.

$$e_a = e_{a,max} \sin(\omega t + \alpha) \quad (2)$$

$$e_b = e_{b,max} \sin(\omega t - \frac{2\pi}{3} + \alpha) \quad (3)$$

$$e_c = e_{c,max} \sin(\omega t + \frac{2\pi}{3} + \alpha) \quad (4)$$

$$v_{J1} = e_a - L_s \frac{di_a}{dt} - R_s i_a \quad (5)$$

$$v_{J2} = e_b - L_s \frac{di_b}{dt} - R_s i_b \quad (6)$$

$$v_{J3} = e_c - L_s \frac{di_c}{dt} - R_s i_c \quad (7)$$

$$v_{J1} = v_{Top} + v(i_{D1}) \quad (8)$$

$$v_{J2} = v_{Top} + v(i_{D3}) \quad (9)$$

$$v_{J3} = v_{Top} + v(i_{D5}) \quad (10)$$

$$v_{Bottom} = v_{J1} + v(i_{D4}) \quad (11)$$

$$v_{Bottom} = v_{J2} + v(i_{D6}) \quad (12)$$

$$v_{Bottom} = v_{J3} + v(i_{D2}) \quad (13)$$

$$i_d = (i_{D1} + i_{D3} + i_{D5}) - C_f \frac{d(v_{Top} - v_{Bottom})}{dt} \quad (14)$$

$$(v_P - v_{Bottom}) = i_d R_{load} + L_{load} \frac{di_d}{dt} \quad (15)$$

$$i_d = (i_{D2} + i_{D4} + i_{D6}) - C_f \frac{d(v_{Top} - v_{Bottom})}{dt} \quad (16)$$

$$i_a + i_b + i_c = 0 \quad (17)$$

$$v_P = v_{Top} - L_f \frac{di_d}{dt} \quad (18)$$

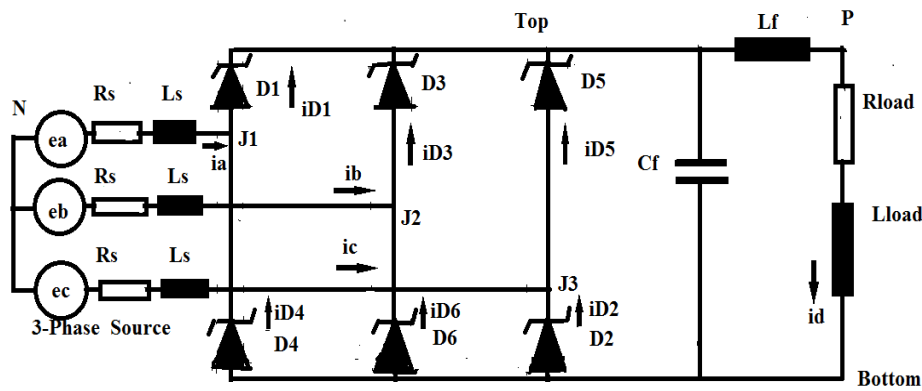


Figure 1. The assumed six-pulse circuit equipped with Zener diodes.

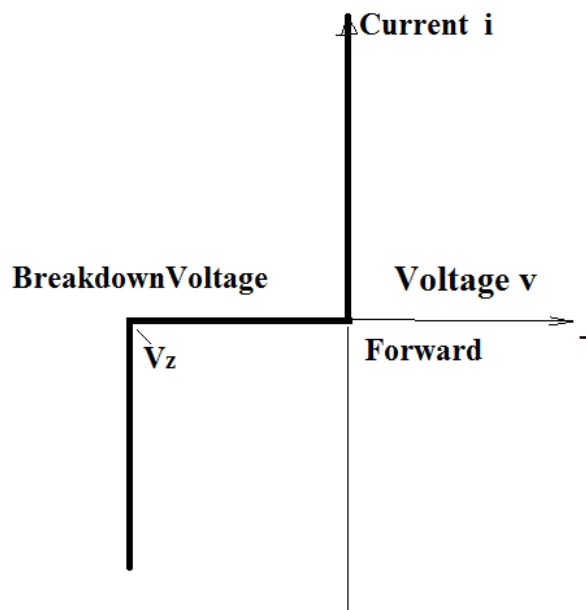


Figure 2. The idealized representation of a Zener diode.

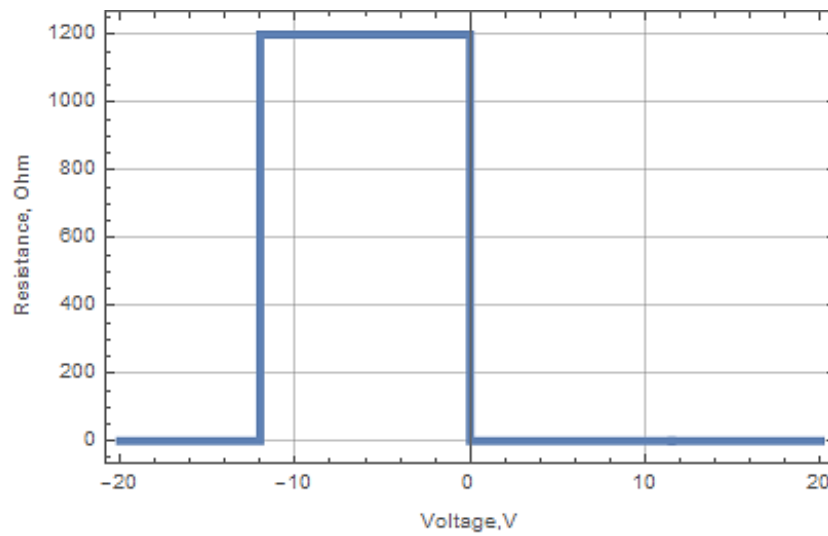


Figure 3. An example for the dependence of the Zener diode's resistance on the voltage, $V_z=12\text{Volts}$ and $r_r=1200\Omega$.

SAMPLE RESULTS

The investigated case studies have the following assumed circuit parameters:

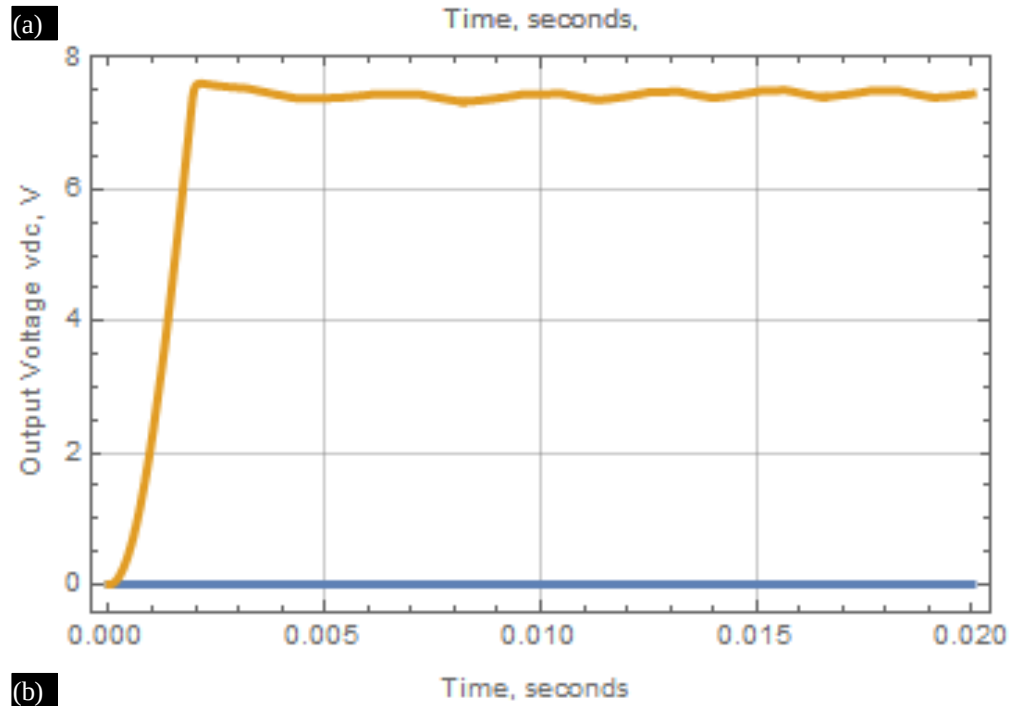
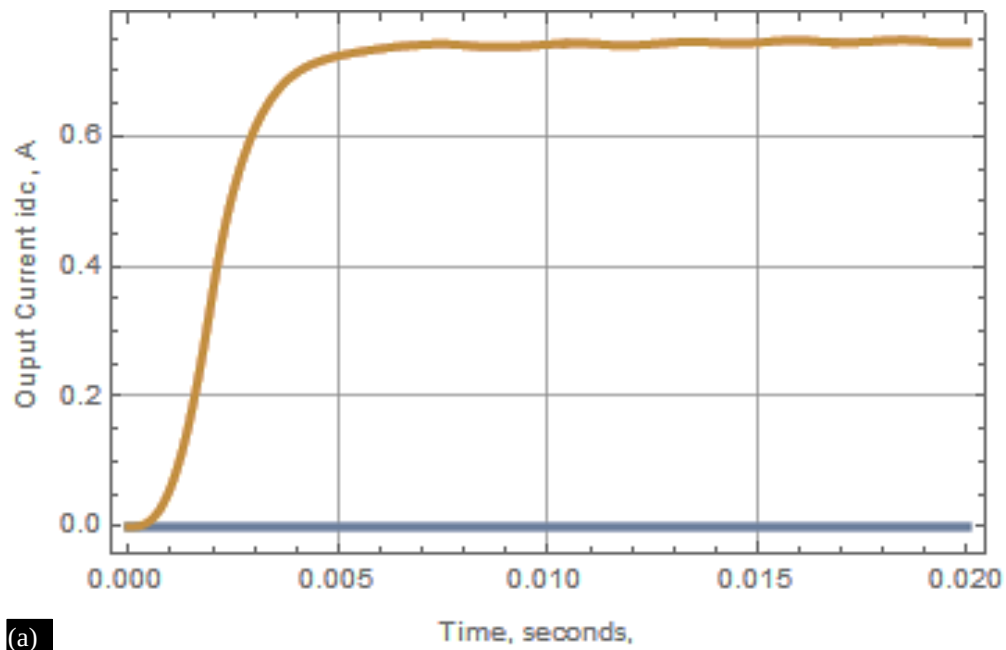
$r_f=10^{-4}\text{ Ohm};$	(resistance in the forward direction)
$r_r=10000\text{ Ohm};$	(resistance in the reverse direction)
$indload=0.01\text{ H};$	(load inductance)
$rload=10\text{ Ohm};$	(load resistance)
$sourceind=0.01\text{H};$	(source inductance)
$sourceresist=0.001\text{ Ohm};$	(source resistance)
$tend=0.02\text{ seconds};$	(solution time range)
$cf=200*10^{-6}\text{ Farad};$	(filter capacitance)
$lfilter=0.001\text{H};$	(filter inductance)
$rfilter=1\text{ Ohm};$	(filter resistance)
$emaxa=12\text{ Volts};$	(peak value of phase a of the source voltage)
$emaxb=12\text{ Volts};$	(peak value of phase b of the source voltage)
$emaxc=12\text{ Volts};$	(peak value of phase c of the source voltage)
$alpha=0\text{ radians}$	(initial phase angle of source voltage of phase a)
$V_z=9\text{ Volts}$	(the Zener voltage)

CASE STUDY A

The Bridge is Composed of Six Zener Diodes:

The plot in Figure 4(a) illustrates the output current i_d as a function of time during the time range $0 \leq t \leq 0.02\text{s}$. It starts from the initial value zero and increases relatively rapidly to its final asymptotic value of about 0.77A after approximately 5 ms . The corresponding plot of the output voltage (v_P-v_{Bottom}) is depicted in Figure 4-(b). It reaches its final value of approximately 7.5V within 2 ms .

The three currents i_a, i_b, i_c flowing through the source terminals are illustrated in Figure 4 (c). They are relatively smooth curves oscillating between -3.5 and $+3.5\text{A}$ and exhibiting small high frequency components as well as a noticeable DC offset. They have to be compared with the strong harmonic polluted diode currents, such as i_{D2}, i_{D4}, i_{D6} shown in Figure 4(d). The indicated three diode currents are seen to vary within the limits -1.7A and $+1.7\text{A}$.



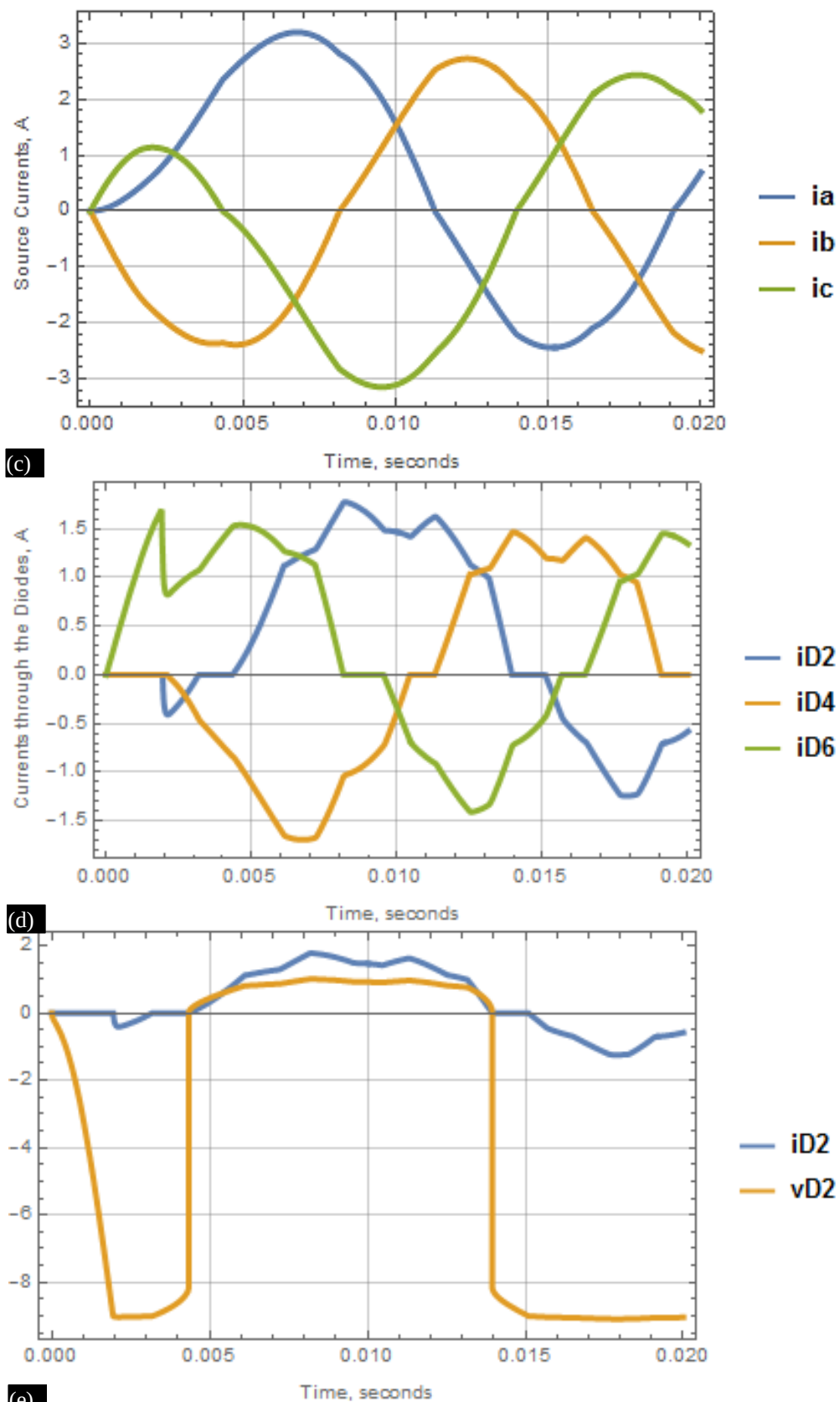


Figure 4. Simulation results of the voltages and currents in the Zener Diode bridge circuit over the time range $0 \leq t \leq 0.02$ s.

Case Study B

The Bridge is Composed of Six Semi-Conductor Diodes:

This case is easily simulated using the same *Mathematica* code (appendix) applied to the above-mentioned case A, but after assuming a very large value $V_z = \text{Infinity}$ for the Zener voltage. The results are depicted in Figure 5. They agree with the plots documented in reference [7] addressing the energization transients of 6-pulse rectifier circuits. This can be considered as a validation of the presented model and program.

The current through and the voltage across the semi-conductor diode D2 are depicted in Figure 6. It is noticed that the reverse voltage increases from approximately 8.5V in Case A to about 17.5V in Case B. For more details on the *Mathematica* code, the references [8-10] should be consulted.

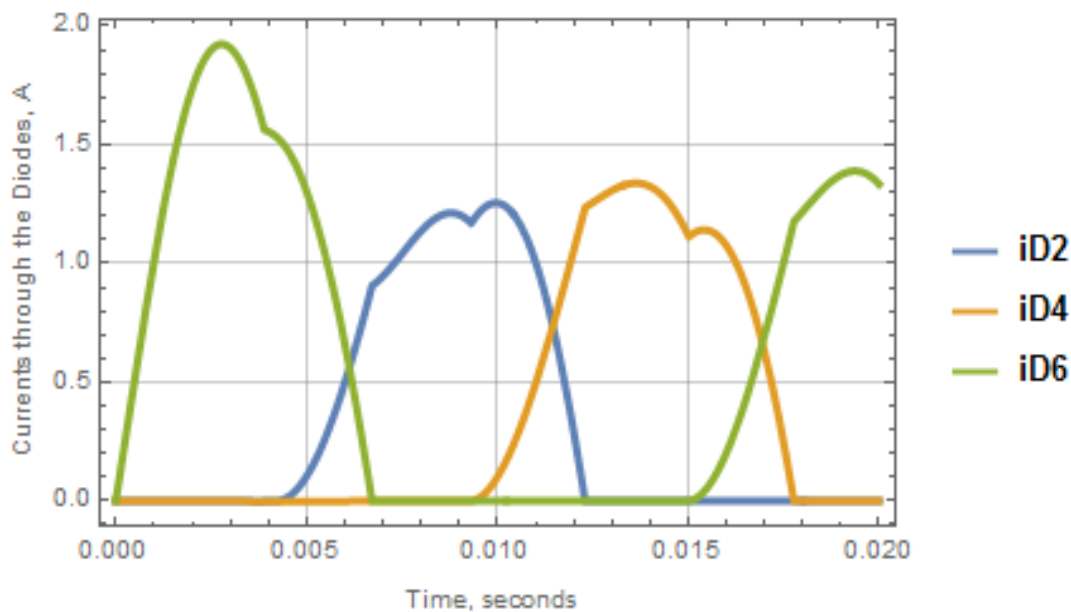


Figure 5. The diode currents of, (special case of $V_z = \text{Infinity}$).

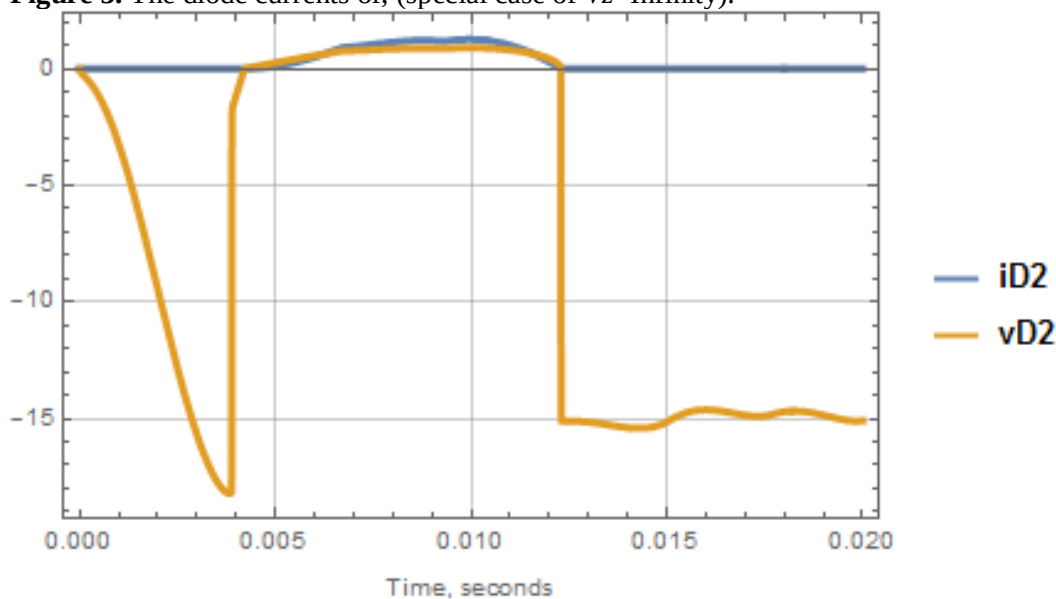


Figure 6. The current through and voltage across the diode D2, (special case of $V_z = \text{Infinity}$).

CONCLUSIONS

A model for the analysis of six-pulse bridge circuits connected to three-phase voltage sources is presented. These circuits can be composed of Zener and/or semi-conductor diodes which are not necessarily identical. A set of 18 simultaneous differential/algebraic equations is derived and programmed in the *Mathematica* environment. The code has control over the required accuracy level and the time range of displaying the results that include the currents flowing through the source terminals, the filter circuit elements and through the six diodes. Two representative case studies are investigated in details, together with a discussion on the validation of the suggested model and its program. Due to its flexibility and ease of implementation, it is believed that this contribution can be of great help to the designers, researchers and practicing engineers of electronic circuits.

REFERENCES

1. Wikipedia (12 March 2022)". Energize Diode". [Online]. Available from https://en.wikipedia.org/wiki/Energize_diode
2. Mohamed Saied. "Novel Technique for The Analysis of Electronic Circuits Including Zener Diodes". Journal of VLSI Design Tools & Technology. 2021; 11(3): 31-41.
3. Vedantu. "Why is Zener diode not used in rectifiers?" [Online]. Available form <https://www.vedantu.com/question-answer/zener-diode-not-used-in-rectifiers-class-12-physics-cbse-60e0858e2483f67e953121a4>
4. Byjus. "Why Is Zener Diode Is Not Used in Rectifiers" [Online]. Available from <https://byjus.com/jee-questions/why-is-zener-diode-is-not-used-in-rectifiers/>
5. Haris Rosdianto. "Energize diode vs rectifier diode: The comparison of Gaussian probability distribution charts from full-wave rectifier". Journal of Physics: Conference Series. 1204(2019): 1-6. (PDF) *Energize diode vs rectifier diode: The comparison of Gaussian probability distribution charts from full-wave rectifier circuits*. Available from: https://www.researchgate.net/publication/332927802_Zener_diode_vs_rectifier_diode_The_comparison_of_Gaussian_probability_distribution_charts_from_full-wave_rectifier_circuits [accessed May 09 2022].
6. eTechnophiles (2021). "Difference between Zener diode and normal Rectifier diode". [Online] Available from <https://www.etechnophiles.com/difference-between-zener-diode-and-normal-rectifier-diode>
7. Mohamed Saied. "The Energization Transients of 6-Pulse Rectifier Circuits". Journal of Power Electronics & Power Systems. July 2020; 10(1): 18-30.
8. Mohamed M. Saied. "Transient Analysis of Electronic Circuits Including Zener Diodes". Journal of VLSI Design Tools & Technology. 2021; 11(3): 31-41.
9. Wolfram Mathematica Tutorial Collection. "Advanced Numerical Differential Equation Solving" in Mathematica, Book, Wolfram Research, 2008, available at <HTTP://WWW.wolfram.com/entertaining/collectivization/AdvancedNumericalDifferentialEquationSolvingInMathematica/>
10. Mohamed Saied. "Harmonic Analysis of 6-Pulse Diode Bridge Rectifier Circuits". Journal of Power Electronics & Power Systems. 2020; 10(2): 1-10

Appendix

THE MATHEMATICA PROGRAM

```
rf=0.00010 ;
rr=10000 ;
indload=0.01;
rload=10;
sourceind=0.01 ;
sourceresist=0.001;
tend=0.05;
```

```

cf= 200*10^-6;
lfilter=0.001;
rfilter= 1;
emaxa=12;
emaxb=12;
emaxc=12;
alpha=0. ;
omega=377;
u[x_]:=1/2+(1/Pi)*ArcTan[ 600 x]/N;
Plot[u[x]/N,{x,-10,10}]/N
ea[t_]:=emaxa*Sin[omega*t+alpha]/N;
eb[t_]:=emaxb*Sin[omega*t+alpha-2*Pi/3]/N;
ec[t_]:=emaxc*Sin[omega*t+alpha+2*Pi/3]/N;
vz=9;
R[v_]:=rf +(rr-rf)*(u[v+vz]-u[v])/N;
sol=
NDSolve[{iDD1[t]\[Equal]iD1[t],
  vj1[t]\[Equal]ea[t]-sourceind*D[ia[t],t]-sourceresist*ia[t],
  vj2[t]\[Equal]eb[t]-sourceind*D[ib[t],t]-sourceresist*ib[t],
  vj3[t]\[Equal]ec[t]-sourceind*D[ic[t],t]-sourceresist*ic[t],
  vj1[t]\[Equal]vTop[t]+ iD1[t]*R[vj1[t]-vTop[t]],
  vj2[t]\[Equal]vTop[t]+ iD3[t]*R[vj2[t]-vTop[t]],
  vj3[t]\[Equal]vTop[t]+ iD5[t]*R[vj3[t]-vTop[t]],
  vBottom[t]\[Equal]vj1[t]+ iD4[t]*R[vBottom[t]-vj1[t]],
  vBottom[t]\[Equal]vj2[t]+ iD6[t]*R[vBottom[t]-vj2[t]],
  vBottom[t]\[Equal]vj3[t]+ iD2[t]*R[vBottom[t]-vj3[t]],
  (iD1[t]+iD3[t]+iD5[t])\[Equal](iD2[t]+iD4[t]+iD6[t]),
  id[t]\[Equal](iD1[t]+iD3[t]+iD5[t])-cf*D[(vTop[t]-vBottom[t]),t],
  vp[t]-vBottom[t]\[Equal]id[t]*rload+indload*D[id[t],t],
  vp[t]\[Equal]vTop[t]-lfilter*D[id[t],t]-rfilter*id[t],
  ia[t]\[Equal]iD1[t]-iD4[t],
  ib[t]\[Equal]iD3[t]-iD6[t],
  ic[t]\[Equal]iD5[t]-iD2[t],
  iDD1[0]\[Equal]0,id[0]\[Equal]0,ia[0]\[Equal]0,ib[0]\[Equal]0,
  ic[0]\[Equal]0,iD1[0]\[Equal]0,iD2[0]\[Equal]0,iD3[0]\[Equal]0,
  iD4[0]\[Equal]0,iD5[0]\[Equal]0,iD6[0]\[Equal]0,vTop[0]\[Equal]0,
  vBottom[0]\[Equal]0,vp[0]\[Equal]0,vj1[0]\[Equal]0,vj2[0]\[Equal]0,
  vj3[0]\[Equal]0},{iDD1,id,ia,ib,ic,iD1,iD2,iD3,iD4,iD5,iD6,vTop,vBottom,
  vp,vj1,vj2,vj3},{t,0,tend},
  Method\[Rule]{EquationSimplification\[Rule]Residual},AccuracyGoal\[Rule]5,
  PrecisionGoal\[Rule]1000 ,MaxSteps\[Rule]Infinity]
Plot[{0,id[t]/.sol},{t,0,tend }, Frame\[Rule]True,
  GridLines\[Rule]Automatic,
  FrameLabel\[Rule]{Time, seconds, (Diode D1 Open-Circuited),
  Ouput Current idc, A}]
Plot[{ ia[t]/.sol, ib[t]/.sol, ic[t]/.sol},{t,0,tend },Frame\[Rule]True,
  GridLines\[Rule]Automatic,
  FrameLabel\[Rule]{Time, seconds , Source Currents, A},
  PlotLegends\[Rule]Automatic,PlotStyle\[Rule]{Thickness[0.01]}]
Plot[{ iD2[t]/.sol, iD4[t]/.sol, iD6[t]/.sol},{t,0,tend }, Frame\[Rule]True,
  GridLines\[Rule]Automatic,
  FrameLabel\[Rule]{Time, seconds ,Currents through the Diodes, A},

```

```

PlotLegends\Rule]Automatic,PlotStyle\Rule]{Thickness[0.01]}]
Plot[{ iD2[t]/.sol, iD4[t]/.sol, iD6[t]/.sol},{t,0,tend }, Frame\Rule]True,
GridLines\Rule]Automatic,
FrameLabel\Rule]{Time, seconds ,Currents through the Diodes, A},
PlotLegends\Rule]Automatic,PlotStyle\Rule]{Thickness[0.01]}]
Plot[{0,(vp[t]-vBottom[t])/.sol } ,{t,0,tend }, Frame\Rule]True,
GridLines\Rule]Automatic,
FrameLabel\Rule]{Time, seconds, (Diode D1 Open-Circuited),
Output Voltage vdc, V}]
Plot[{0,id[t]/.sol },{t,(tend-1/60),tend}]
Plot[{0,ia[t]/.sol },{t,(tend-1/60),tend} ]
Plot[{0, iD1[t]/.sol},{t,(tend-1/60),tend}]
Plot[{0,(vp[t]-vBottom[t])/.sol } ,{t,(tend-1/60),tend}]

```